FARM MANAGEMENT SYSTEM

Salesforce-Based Agricultural Data Management Application

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- **DEPARTMENT :** B.Sc. Computer Science
- **YEAR :** Final Year

1. INTRODUCTION

The Farm Management System (FMS) is a Salesforce-based application developed to digitize and streamline agricultural operations. It manages farm, crop, farmer, and buyer information in a centralized system. The application uses Salesforce objects, fields, relationships, validation rules, reports, and dashboards to organize agricultural data and support efficient farm management.

2. PROBLEM STATEMENT

Managing farm and agricultural information manually can become difficult when the number of records increases. Farm, crop, farmer, and buyer details may be maintained in registers, paper files, or spreadsheets. This can result in incomplete or incorrect data, data loss, delayed reports, and difficulty in monitoring farm productivity and sales. There is a need for a centralized system that stores agricultural information securely and provides organized reports and dashboards.

3. SOLUTION

The Farm Management System provides a Salesforce-based solution for managing agricultural information in one centralized platform. It stores farm, crop, farmer, and buyer records and connects related data using Salesforce relationships. Validation rules help prevent incorrect entries, while reports and dashboards provide organized information for monitoring and analysis. The system reduces manual paperwork and supports faster and more accurate farm management.

4. OBJECTIVES

To maintain farm information in a centralized system.

To record and manage crop production details.

To maintain farmer and buyer information.

To connect related agricultural records using Salesforce relationships.

To validate important data and reduce incorrect entries.

To generate reports and dashboards for monitoring farm information.

To reduce manual paperwork and improve data accessibility.

5. PHASES OF DEVELOPMENT

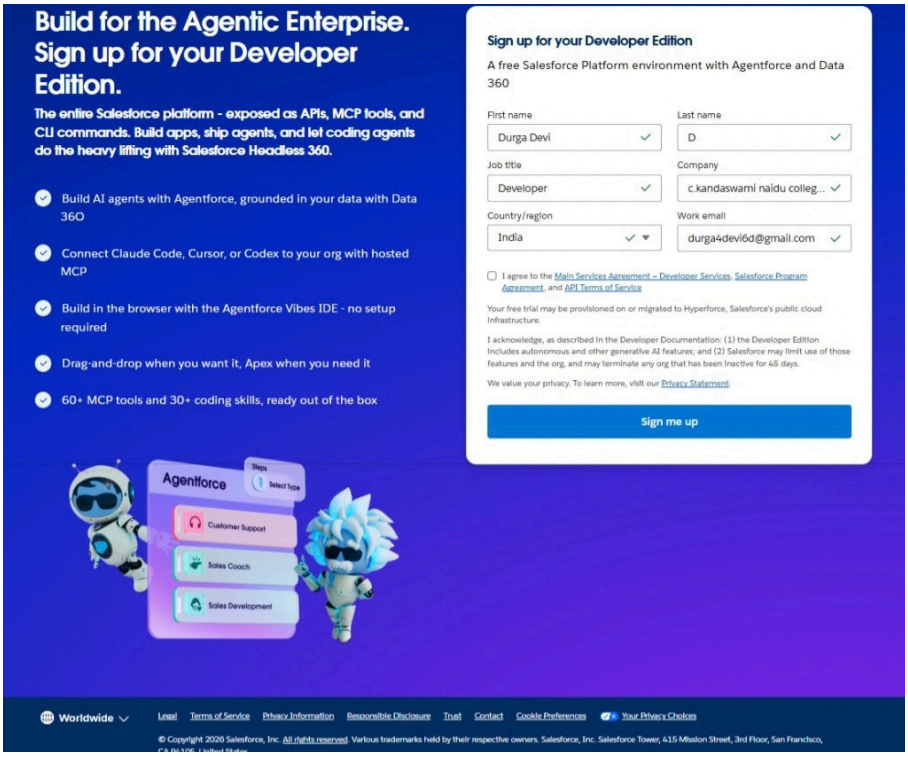
PHASE 1: REQUIREMENT ANALYSIS AND PLANNING

In this phase, the requirements of the Farm Management System were identified. The main requirement was to create a centralized system for storing farm, crop, farmer, and buyer information and for generating reports and dashboards. Salesforce was selected as the development platform. The project was planned using custom objects, fields, relationships, validation rules, reports, and dashboards.

Main users: Farmers, Farm Managers, Buyers, and System Administrators.

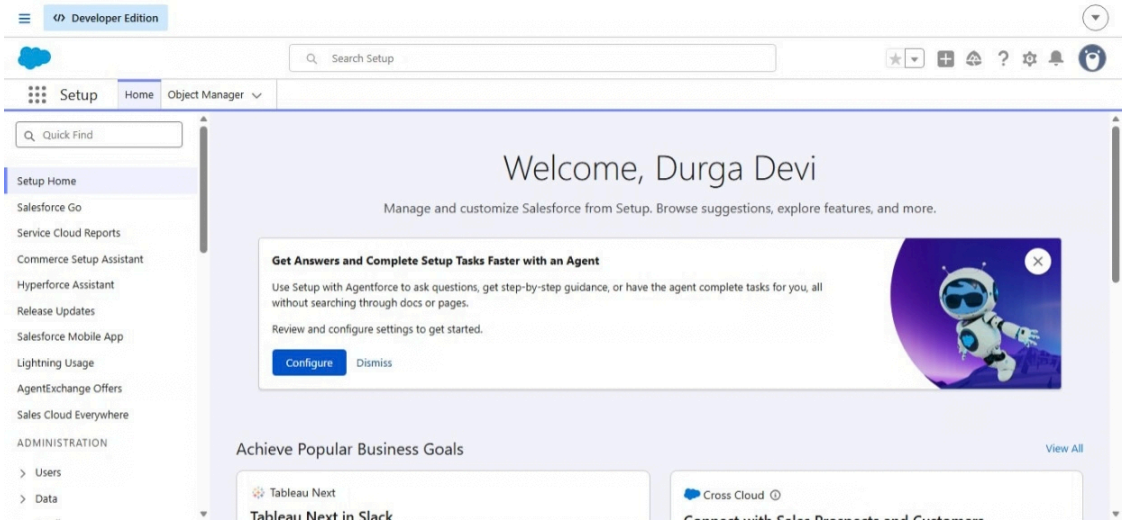
Main data: Farm, Crop, Farmer, and Buyer.

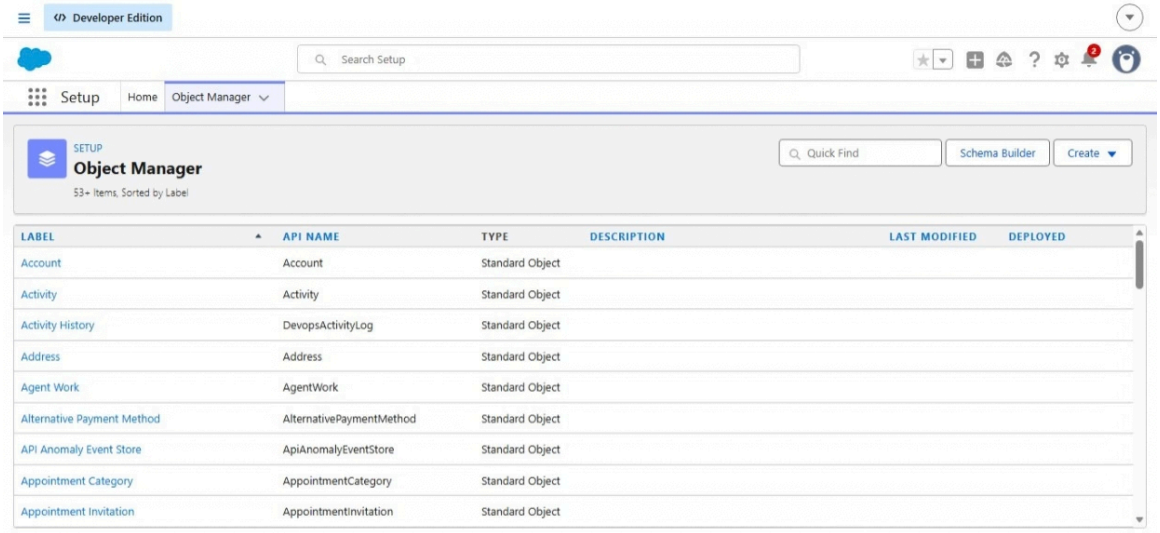
Main outputs: Reports, dashboards, and organized agricultural records.



PHASE 2: SALESFORCE DEVELOPMENT – BACKEND AND CONFIGURATION

The main backend components were created in Salesforce using custom objects, fields, and relationships.





Farm Object

Farm Name

Farm Type

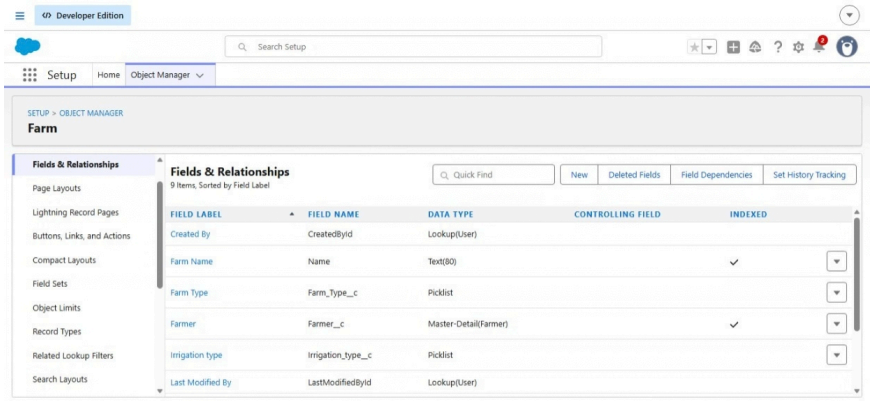
Soil Quality

Irrigation Type

Location

Size

Master-Detail relationship with Farmer



Crop Object

Crop Name

Planting Date

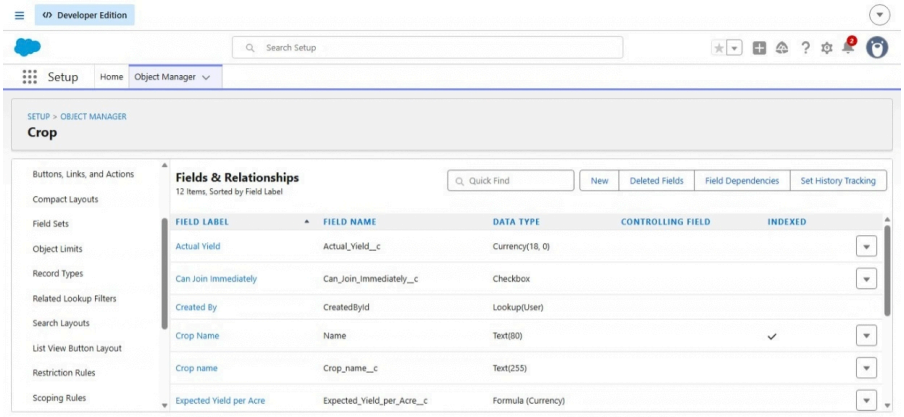
Harvest Date

Farm lookup relationship

Actual Yield

Total Acres Harvested

Expected Yield per Acre formula



Farmer Object

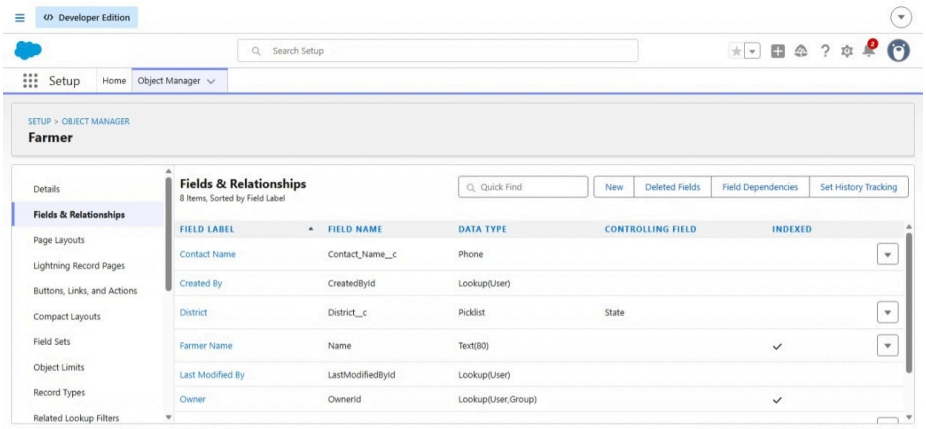
Contact Number

State

District

Village

State → District field dependency



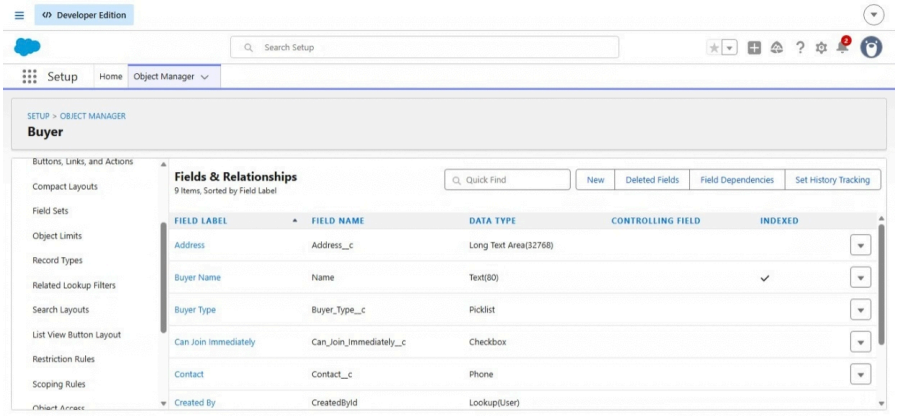
Buyer Object

Contact

Email

Address

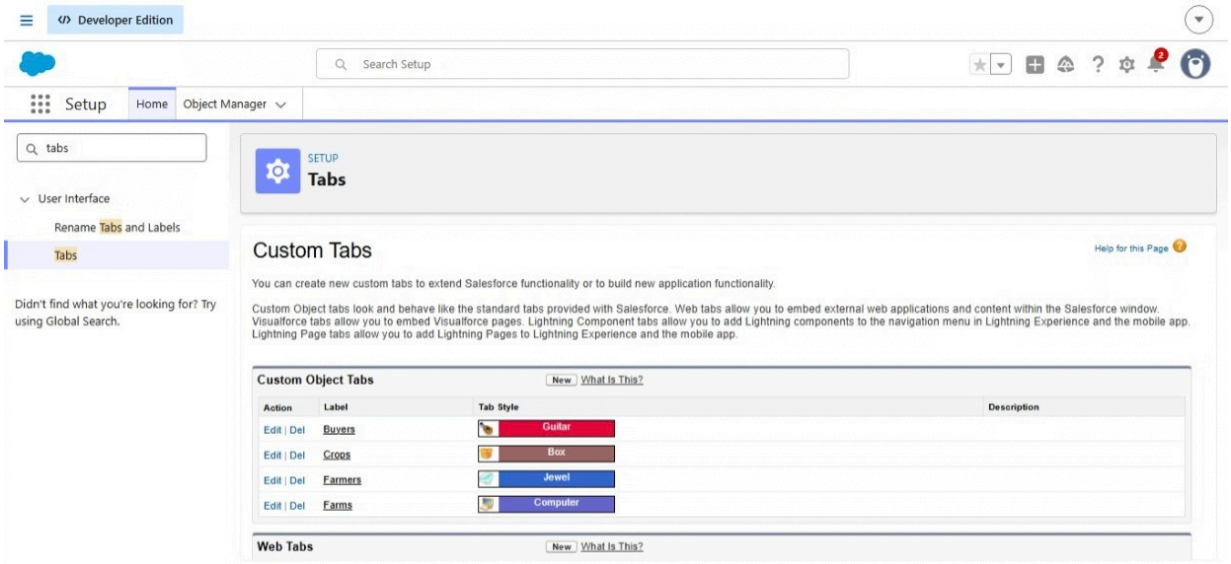
Buyer Type



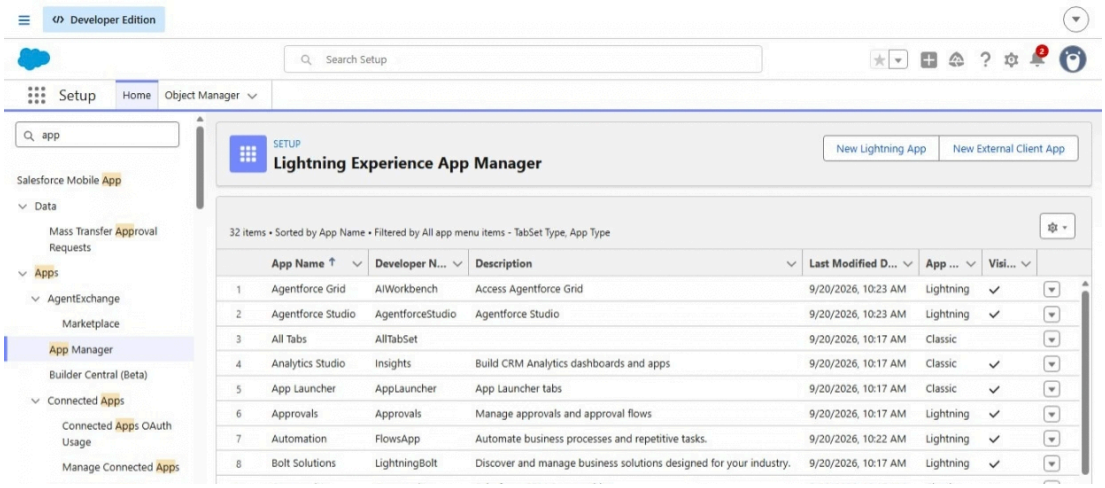
PHASE 3: UI/UX DEVELOPMENT AND CUSTOMIZATION

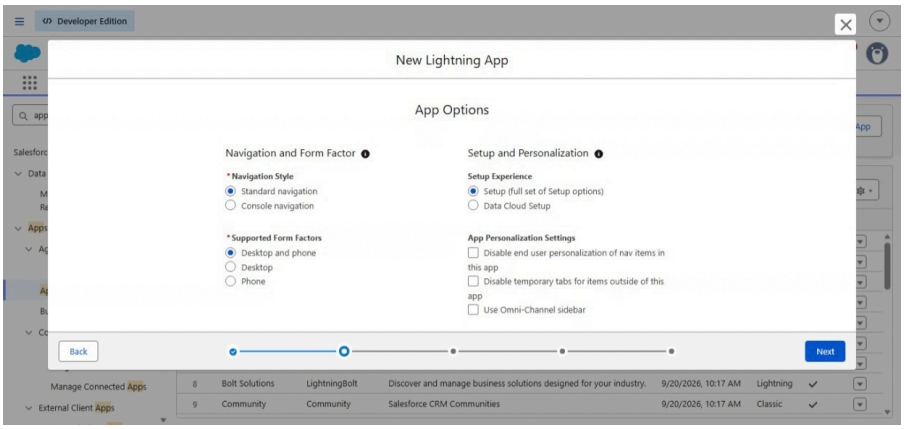
Custom tabs were created for Farm, Crop, Farmer, and Buyer objects. A Lightning App named Farm Management System was configured to provide easy access to the application components. Reports and Dashboards were also added to the application navigation for convenient monitoring.

Custom Tabs: Farm, Crop, Farmer, Buyer

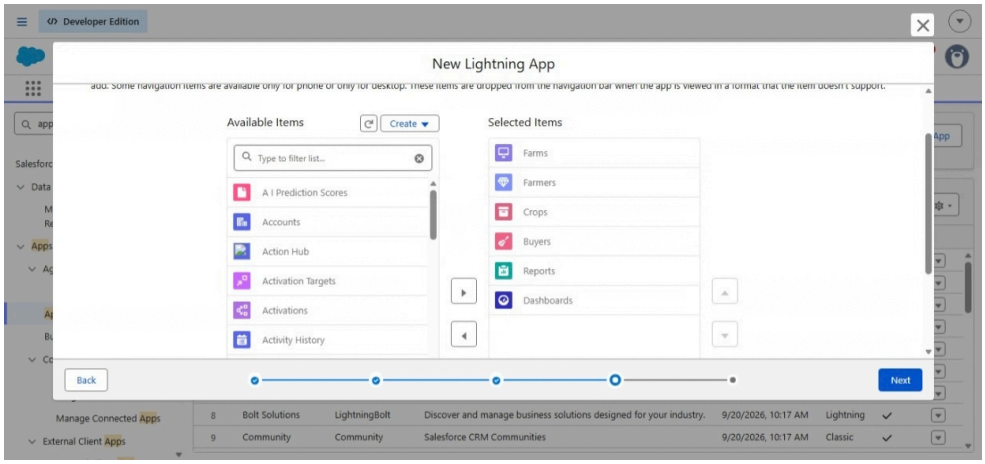


Lightning App: Farm Management System





Navigation: Farm, Crop, Farmer, Buyer, Reports, Dashboards



PHASE 4: DATA MIGRATION, TESTING AND SECURITY

Farm, Crop, Farmer, and Buyer records were entered into Salesforce and the relationships between records were checked. The application was tested by verifying record creation, validation rules, field dependencies, relationships, user access, sharing rules, reports, and dashboards. Salesforce profiles, roles, users, and sharing rules were configured to support controlled access.

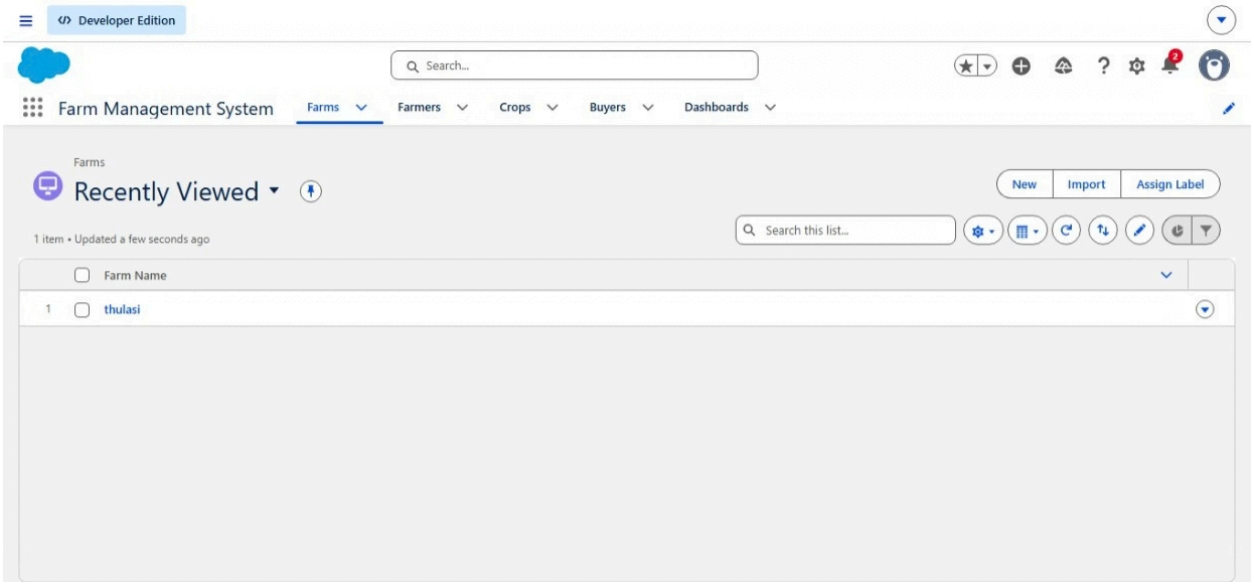
Record creation and data entry testing

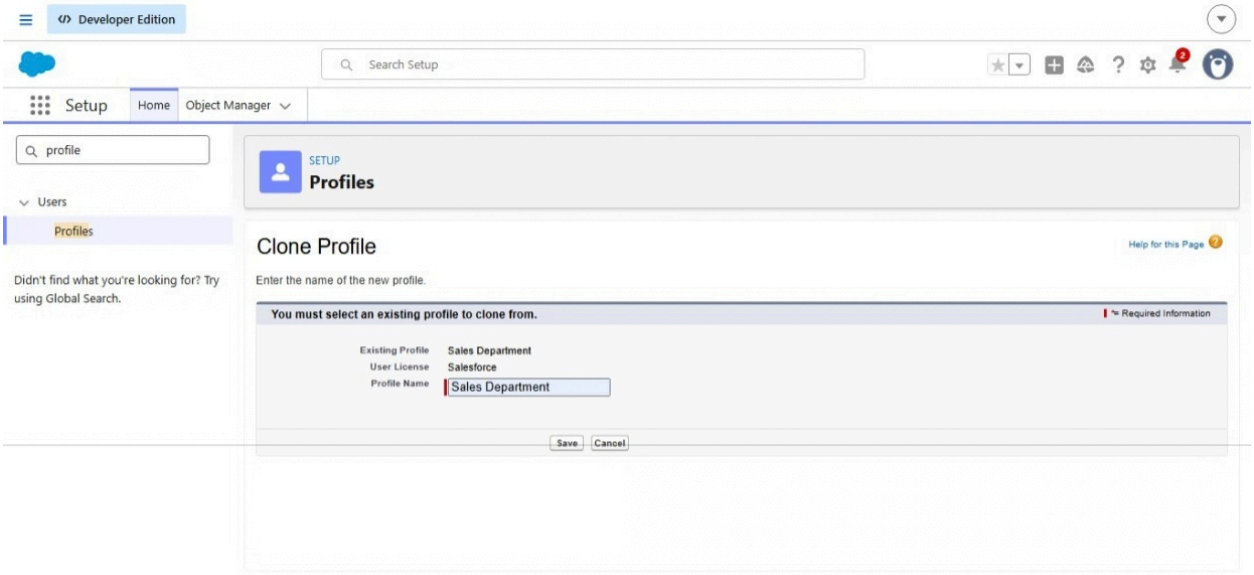
Validation and dependency testing

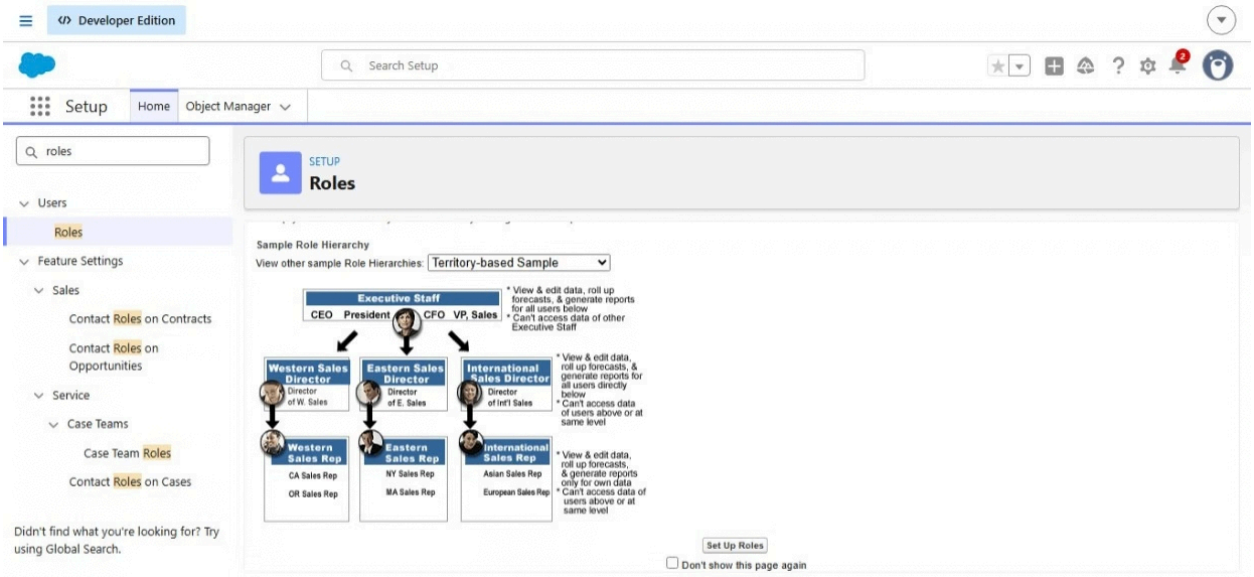
Relationship testing

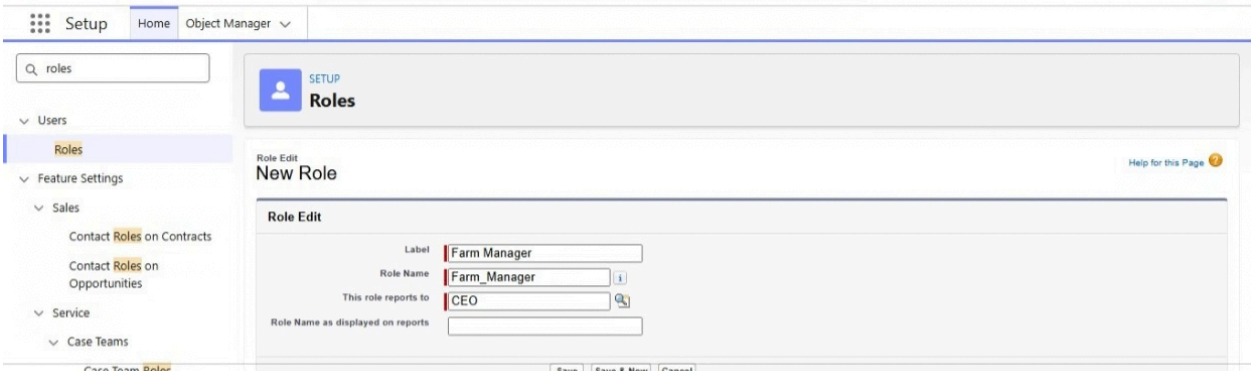
Profile, role, and sharing rule testing

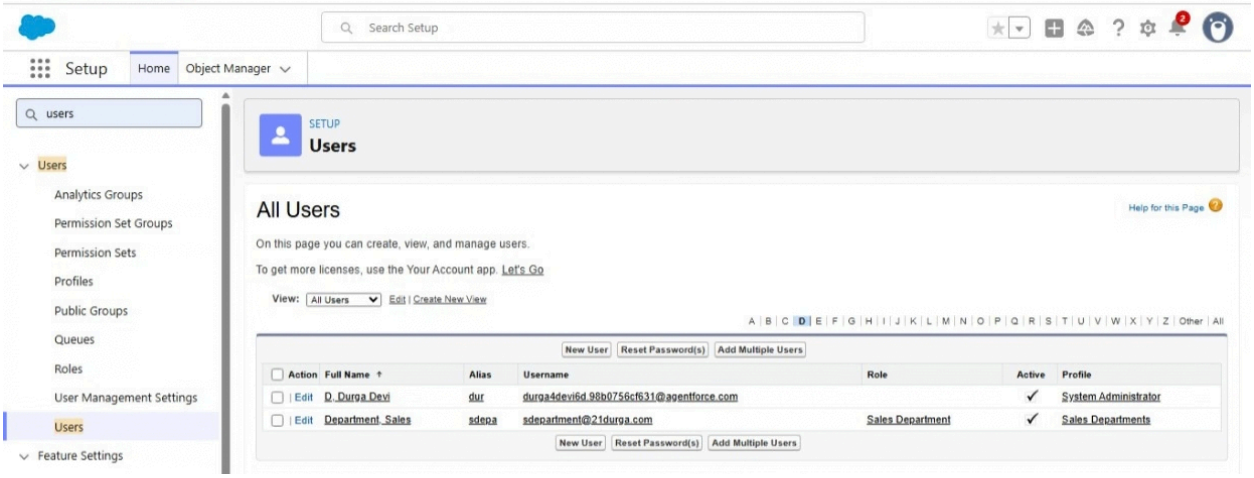
Report and dashboard verification

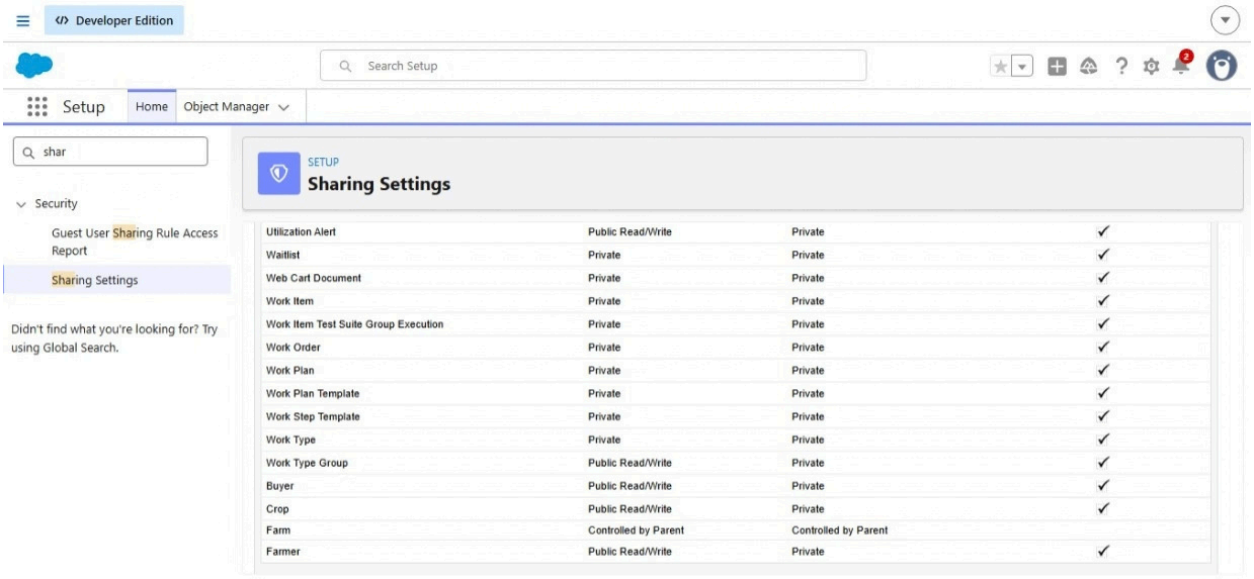


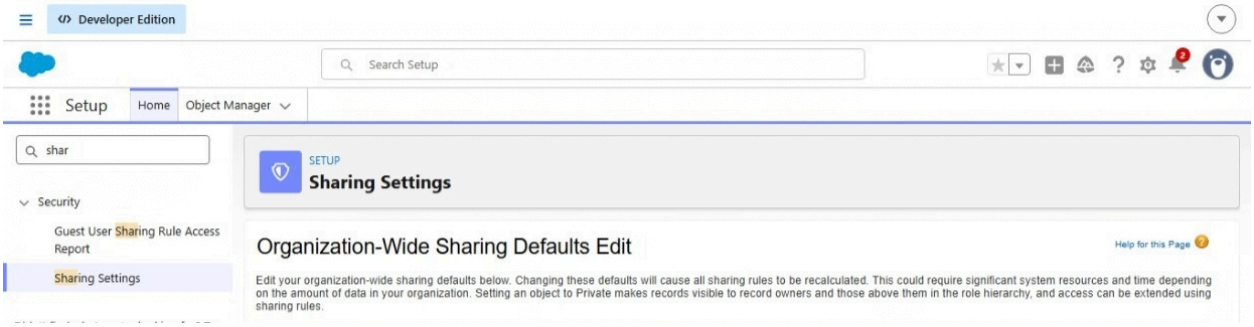



Developer Edition
Search Setup
Setup
Home
Object Manager
roles
Users
Roles
Feature Settings
Sales
Contact Roles on Contracts
Contact Roles on Opportunities
Service
Case Teams
Case Team Roles
Contact Roles on Cases
Didn't find what you're looking for? Try using Global Search.
SETUP
Roles
Sample Role Hierarchy
View other sample Role Hierarchies: Territory-based Sample
Executive Staff
CEO President
CFO VP, Sales
* View & edit data, roll up forecasts, & generate reports for all users below
* Can't access data of other Executive Staff
Western Sales Director
Director of W. Sales
Eastern Sales Director
Director of E. Sales
International Sales Director
Director of Int'l Sales
* View & edit data, roll up forecasts, & generate reports for all users directly below
* Can't access data of users above or at same level
Western Sales Rep
CA Sales Rep
OR Sales Rep
Eastern Sales Rep
NY Sales Rep
MA Sales Rep
International Sales Rep
Asian Sales Rep
European Sales Rep
* View & edit data, roll up forecasts, & generate reports only for own data
* Can't access data of users above or at same level
Set Up Roles
Don't show this page again


Setup
Home
Object Manager
roles
Users
Roles
Feature Settings
Sales
Contact Roles on Contracts
Contact Roles on Opportunities
Service
Case Teams
SETUP
Roles
Role Edit
New Role
Help for this Page
Role Edit
Label
Farm Manager
Role Name
Farm_Manager
This role reports to
CEO
Role Name as displayed on reports


Search Setup
Setup
Home
Object Manager
users
Users
Analytics Groups
Permission Set Groups
Permission Sets
Profiles
Public Groups
Queues
Roles
User Management Settings
Users
Feature Settings
SETUP
Users
All Users
Help for this Page
On this page you can create, view, and manage users.
To get more licenses, use the Your Account app. Let's Go
View: All Users
Edit | Create New View
A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All
New User
Reset Password(s)
Add Multiple Users
Action
Full Name
Alias
Username
Role
Active
Profile
Edit
D. Durga Devi
dur
durga4devi6d 98b0756cf631@agentforce.com
System Administrator
Edit
Department, Sales
sdepa
sdepartment@21durga.com
Sales Department
Sales Departments
New User
Reset Password(s)
Add Multiple Users





PHASE 5: DEPLOYMENT, DOCUMENTATION AND MAINTENANCE

The Farm Management System was completed and configured in the Salesforce environment. The final application was checked to ensure that the main features were functioning correctly. Reports and dashboards were used to view summarized agricultural information. The system can be maintained in the future by adding new farms, crops, farmers, buyers, reports, and additional features.

6. ADVANTAGES AND DISADVANTAGES

Advantages

Centralized agricultural data.

Reduced paperwork and manual work.

Improved data accuracy and accessibility.

Easy management of farm, crop, farmer, and buyer records.

Reports and dashboards support monitoring and analysis.

Can be extended with additional features.

Disadvantages

Requires Salesforce access.

Internet connection is generally required.

Limited offline access.

Depends on the Salesforce platform and its configuration.

7. CONCLUSION

The Farm Management System was successfully developed using Salesforce to organize and manage agricultural information. It provides a centralized system for farm, crop, farmer, and buyer records and uses Salesforce features such as custom objects, fields, relationships, validation rules, reports, and dashboards. The project helps reduce manual record keeping, improve data accessibility, and support organized monitoring of farm information.

8. FUTURE SCOPE

Mobile application integration.

AI-based crop prediction.

Weather data integration.

IoT sensor connectivity.

Government scheme integration.

Market price forecasting.

9. APPENDIX

Source: Salesforce configuration and metadata.

Dataset: Farm, Crop, Farmer, and Buyer records.

Tools Used: Salesforce Developer Org.

Thank You